Grade 6 Accelerated Name____________________
Unit 9 Date____________________
Lesson 1 Practice Period____________________

For questions 1-6, complete the table below.

	Data Description	**Examples**	**Data Classification**
1.	Scores on a Math test	$70, 76, 78, 84, 92, 97$	
2.	Favorite classes taken by a student survey	Math, Science, Social Studies, Language Arts, Physical Education	
3.			numerical
4.			categorical
5.	Sports played after school		
6.	Number of students in each class period		

7. __________________ data is a type of data that is divided into groups. This type of data is qualitative.

For questions 8-10, classify each as Categorical or Numerical.

8. Number of teachers in each department at a large high school.

9. Eye colors of all the students in your class.

10. Height of your family members.

Grade 6 Accelerated Name____________________
Unit 9 Date____________________
Lesson 2 Practice Period____________________

For questions 1-7, determine whether each question is statistical or not statistical.

1. What Ocean does Florida border?

2. What size jerseys will each team at the school need?

3. How many pets do you have?

4. Is Orlando the capital of Florida?

5. How many followers do your classmates have on Twitter?

6. What is each member of your class's favorite movie?

7. The question “Who was the first President of the United States?” is not statistical. Use the definition to justify why it is not a statistical question.

8. Decide if the question, “How many houses are in your neighborhood?” is statistical or non-statistical. Explain your decision.

9. Create an example of a statistical question.

10. Create a counterexample that is NOT a statistical question.

Grade 6 Accelerated Name____________________
Unit 9 Date____________________
Lesson 3 Practice Period____________________

Disney World in Orlando Florida opened Magic Kingdom in 1971. Since then, it has become a world-renowned attraction. Use the table that shows the number of visitors recorded (in millions) for questions 1-3.

Year	**Total Visitors (in millions)**
2010	16.972
2012	17.536
2014	19.332
2016	20.395
2018	20.859
2020	6.941

1. What is the mean attendance during this span of years? Round to the hundredths place.

2. Interpret the mean attendance for the 6 years.

3. Explain why the mean in the data above does not give information about Magic Kingdom's visitor attendance today?

The 5 deepest lakes in the United States by depth are listed in the table below. Use the data for questions 4-5.

Lake	**Total Depth (in feet)**
Crater Lake	1,949
Lake Tahoe	1,645
Lake Chelan	1,486
Lake Superior	1,332
Lake Pend Oreille	1,150

4. Find the mean of the data.

5. Interpret the mean depth of the lakes based on the data.

The Tampa Bay Rays had 23 pitchers in their 2021 season. A random sample of 7 pitchers are listed in the table. Use the data for questions 6-7.

Pitchers Name	Pitchers Age
Corey Kluber	35
Luis Patino	22
Brooks Raley	33
Shane Baz	22
Nick Anderson	31
Ryan Yargrough	30
Josh Fleming	25

6. Find the mean age of the pitchers given.

7. Interpret the mean based on the data given from the sample of pitchers.

A farmer in northeast Florida raises chickens, pigs, and cows. He sells his chicken eggs to people in surrounding areas. He begins keeping a record of what each chicken lays per week to best estimate the amount he will have for sale from week to week. Use the data in the table for questions 8-10.

Eggs Laid by Chickens in One Week			
Chicken A:	6	**Chicken E:**	6
Chicken B:	5	**Chicken F:**	7
Chicken C:	9	**Chicken G:**	8
Chicken D:	3	**Chicken H:**	6

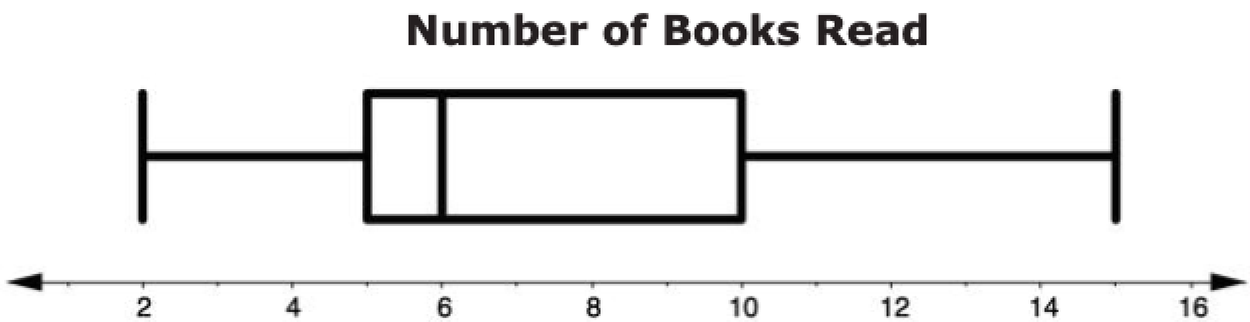
8. What number of eggs do most chickens lay at this farm?

9. What is the median amount of eggs produced?

10. What does the value of the median represent in this context?

Grade 6 Accelerated Name____________________
Unit 9 Date____________________
Lesson 4 Practice Period____________________

A survey of a 6^{th} grade class was given asking what number of books students read in the last two months of school. The data in the box plot shows the results. Use the box plot for questions 1-5.



1. The minimum books read is ____________, while the maximum is ___________.

2. The median number of books read is _______________.

3. The upper quartile is _______________.

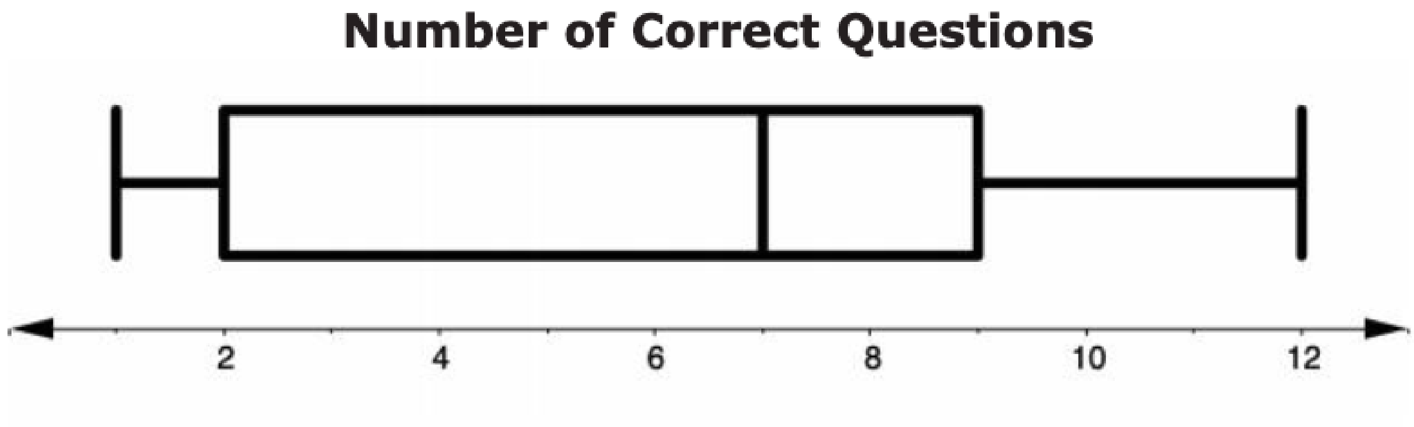
4. The lower quartile is _______________.

5. The range is _______________.

6. The Interquartile range (IQR) is ______________.

7. A box plot divides data into 4 equal parts that each contain ________% of data from a given data set.

A quiz was given to students in a science class. Use the data in the box plot for questions 8-10.



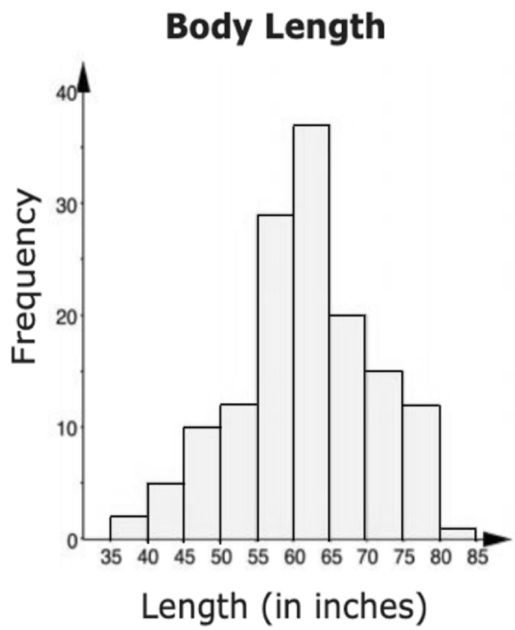
8. What percent of students got between 9 and 12 questions correct?

9. What percent of students got between 2 and 9 questions correct?

10. What percent of students got between 7 and 12 questions correct?

Grade 6 Accelerated		Name____________________
Unit 9				Date____________________
Lesson 5 Practice		Period___________________

In a study of wild bears, researchers measured the length, in inches (in.) of 143 wild bears that ranged in age from newborn to 15 years old. The data was used to make the histogram shown.



1. How many bears are between 45 and 50 inches long?

2. Which single range of length has the most bears?

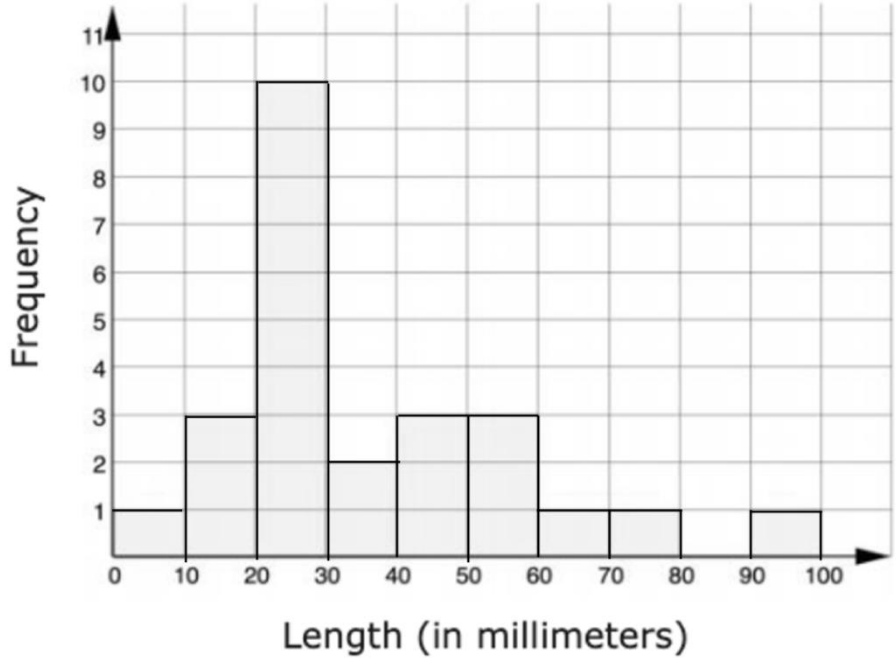
3. What is the smallest possible length of a bear in this data set?

4. Are there more bears measuring less than 65 inches or more than 65 inches?

5. If bears older than 15 were added to the sample, hypothesize what would happen to the spread of the data.

6. Mark an "M" above the bar that contains the median.

An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. Lengths of earthworms provide information about their ages. The farmer measured the lengths of earthworms in one container. The farmer measured the earthworms in millimeters then created a histogram from the data.



7. Is it possible for the sample to contain an earthworm measuring more than 100 mm?

8. More than half of the earthworms measure less than _________ millimeters.

9. How many data pieces are in the histogram?

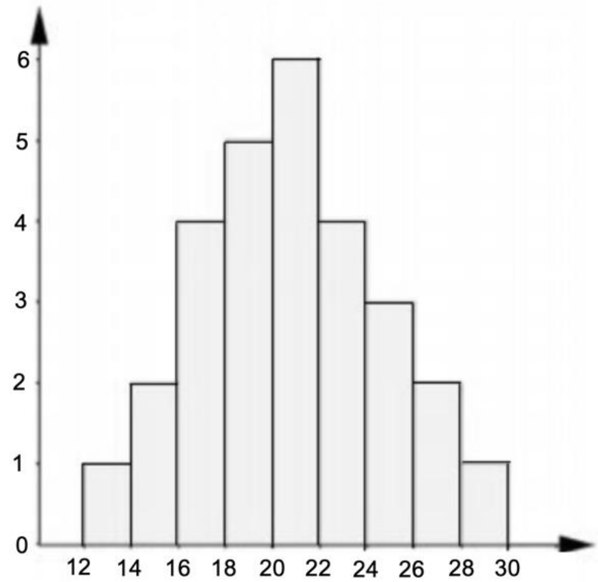
10. What single range of lengths has the most earthworms?

Grade 6 Accelerated Name____________________
Unit 9 Date____________________
Lesson 6 Practice Period____________________

1. A _________________ skewed graph has a longer trail of data to the right and the peak of the data on the left.

2. The shape of the histogram is

- ○ symmetric.
- ○ left skewed.
- ○ right skewed.



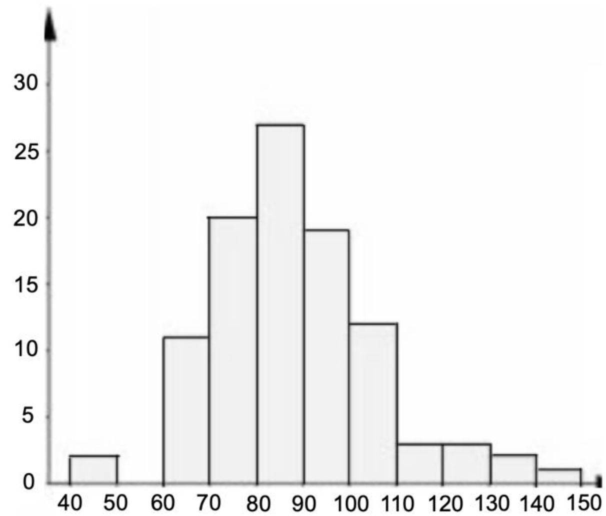
3. If the data value _________ were added to the histogram above, it would be considered an outlier.

4. Sketch a histogram whose shape can be described as right skewed.

5. Sketch a histogram whose shape is not symmetrical and has more than one peak.

6. Suppose a histogram of a data set is symmetrical. If an outlier is added to a data set that is far greater than the median, which direction will the graph be skewed?

Use the histogram for questions 7-10.



7. The shape of the histogram can be described as

- symmetric.
- left skewed.
- right skewed.

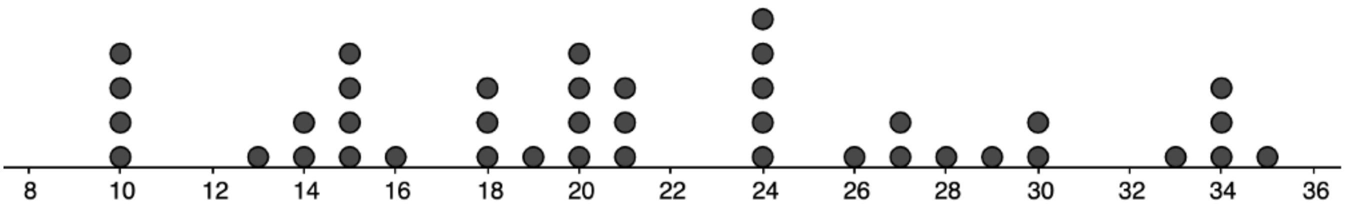
8. Is there a gap in the histogram? If so, where?

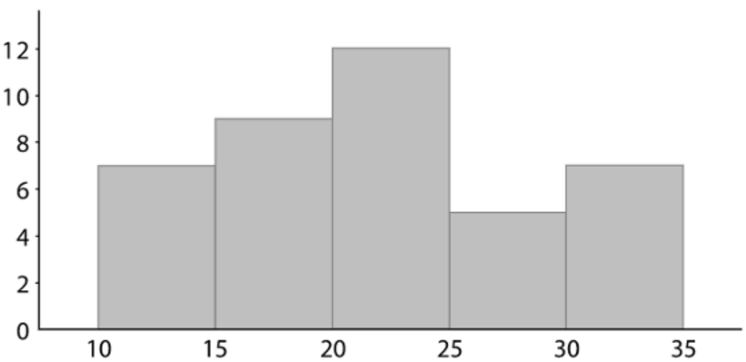
9. Is there a cluster in the histogram? If so, where?

10. If the data value __________ were added to the histogram above, it would be considered an outlier.

Grade 6 Accelerated Name____________________
Unit 9 Date____________________
Lesson 7 Practice Period____________________

Here is a histogram and dot plot showing the weights, in pounds, of 40 dogs at a dog show.





Dog weights in pounds

1. Which data display can be used to determine the mean weight?

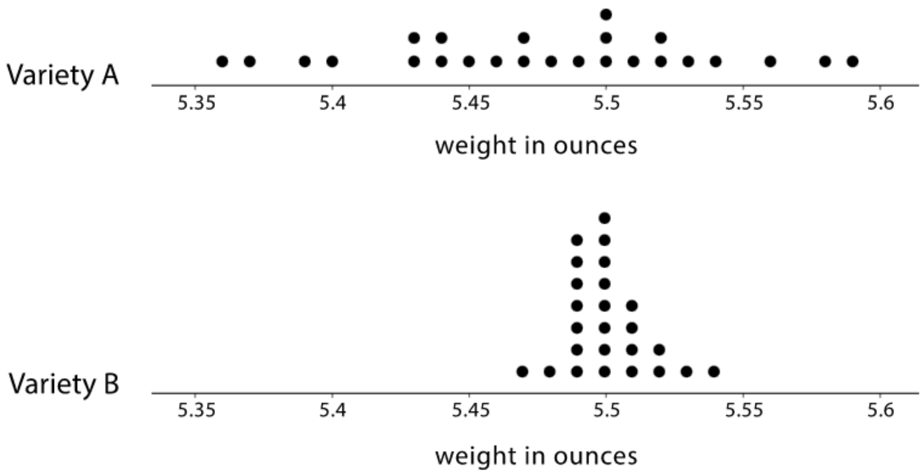
2. Find the mean weight. Interpret the mean.

3. Which data display can be used to determine the median weight?

4. Find the median weight. Interpret the median.

A farmer sells tomatoes in packages of ten. She would like the tomatoes in each package to all be about the same size and close to 5.5 ounces in weight.

The farmer is considering two different tomato varieties: Variety A and Variety B. The dot plots show her data.



5. Describe the shape and distribution of Variety A.

6. Describe the shape and distribution of Variety B.

7. What percentage of tomatoes in Variety A were less than 5.49 ounces?

8. What percentage of tomatoes in Variety B were less than 5.49 ounces?

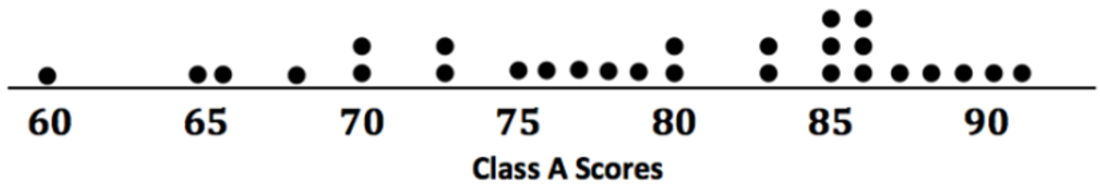
9. What is the range of weights in Variety A and B?

Variety A	Variety B

10. Which tomato variety should the farmer choose? Explain your reasoning.

Grade 6 Accelerated Name____________________
Unit 9 Date____________________
Lesson 8 Practice Period____________________

The grades (based on 100 points) of a class of students is shown in the dot plot.



1. Describe the spread and shape of the distribution.

2. Find and interpret the mean scores. Round to the nearest tenth.

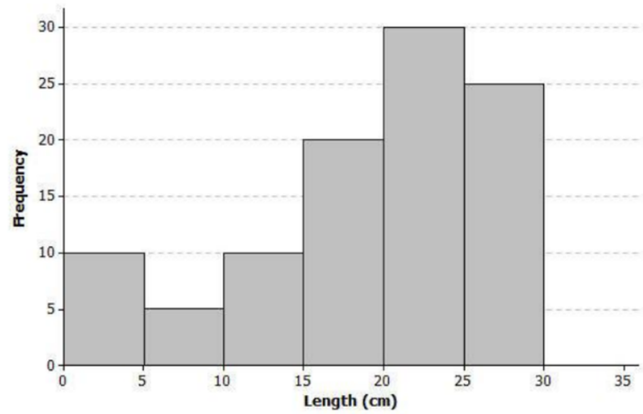
3. Find and interpret the median scores.

4. How would the mean and median change if you found the mean and median of the top 10 grades on the test?

5. Find the mean of the top 10 scores.

6. Find the median of the top 10 scores.

Scientists collected data from many yellow perch because they were concerned about the survival of the yellow perch. Scientists captured yellow perch from a local lake. They recorded data on each fish and then returned each fish to the lake. Consider the following histogram of data on the length (in centimeters) for a sample.

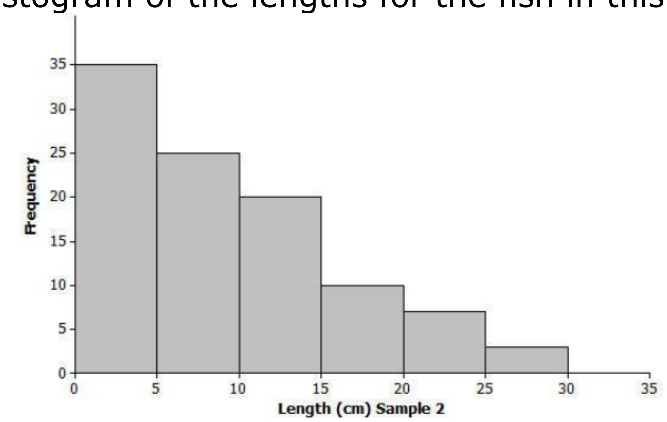


7. Which measure of the center can be determined by the histogram, the mean or the median?

8. Estimate the median length of the yellow perch in the sample.

9. Based on the shape of this data distribution, do you think the mean length of a yellow perch would be greater than, less than, or the same as your estimate of the median?

Another sample of Great Lake yellow perch from a different lake was collected. A histogram of the lengths for the fish in this sample is shown below.



10. Based on the shape of this data distribution, do you think the median length of a yellow perch from this second sample would be greater than, less than, or the same as the median length from the first sample?

Grade 6 Accelerated
Unit 9
Lesson 9 Practice

Name____________________
Date____________________
Period____________________

The number of letters in 20 students' full names is shown below.

$$6, 7, 8, 8, 10, 11, 12, 12, 12, 12, 13, 15, 15, 16, 18, 19, 19, 21, 24, 32$$

1. Determine the five-number summary for the data set.

   Minimum: ___________

   First Quartile: ___________

   Median: ___________

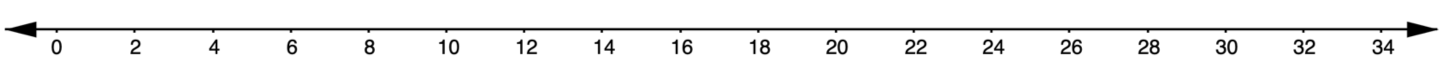
   Third Quartile: ___________

   Maximum: ___________

2. What is the range of the data?

3. What is the interquartile range of the data?

4. Create a box plot of the number of letters in the students' full names.



A school's data on the number of daily absences for the last 14 days are shown.

7	8	11	6	10	4	13	2	5	8	10	13	6	17

5. Determine the five-number summary for the data set.

6. Create a box plot of the daily absences.

7. Explain how you determine the scale to use for the number line of your box plot.

The free-throw percentages of twelve players on a basketball team are shown.

71.2	88.7	50	81.1	84.3	78.7	74.5	82.4	85.1	91.5	76.5	87.2

8. Determine the five-number summary for the data set.

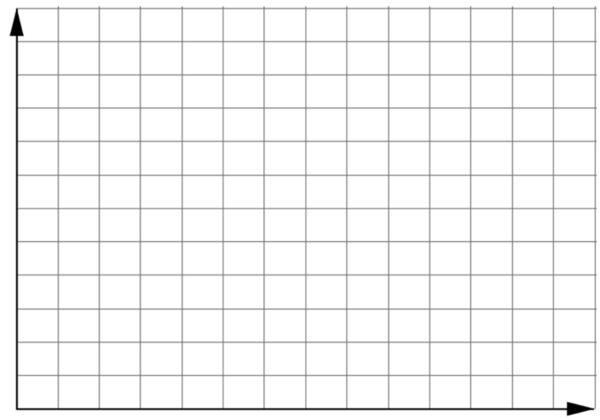
9. Create a box plot of the team's free-throw percentages.

10. What is the interquartile range?

Grade 6 Name____________________
Unit 9 Date____________________
Lesson 10 Practice Period____________________

Arturo asked students at his middle school, how many hours they played video games the week of spring break. The results are summarized in the table.

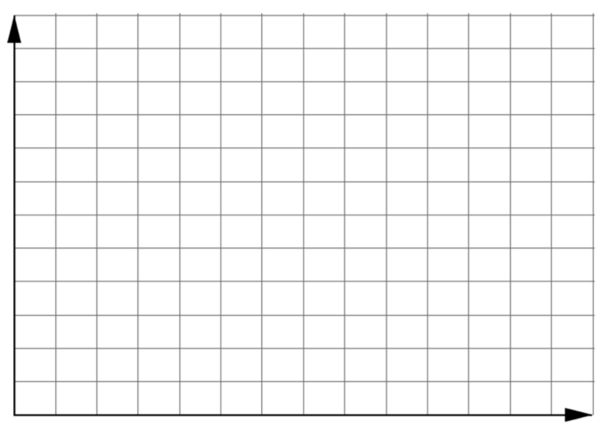
Hours of Video Games	Number of Students
$0-4$	70
$5-9$	90
$10-14$	25
$15-19$	5



1. Create a histogram using the given frequency table.
2. Which interval does the median number of students fall within?

A teacher scored a recent exit ticket and displayed the data on the board.

81	84	57	60	45	92	67
48	78	83	45	75	52	70
95	67	42	62	47	86	48



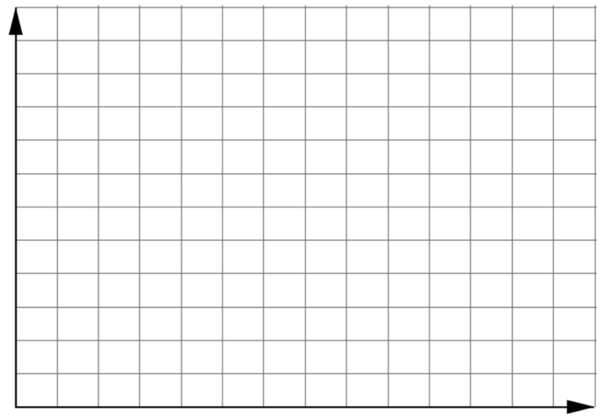
3. Determine the range of the data.
4. Determine the interval width.

5. Create a histogram using the frequency table and interval width in question 4.

6. What was the least common interval of the exit ticket scores?

Ananya wanted to know how long it takes for her 6th grade math class to get to school every morning.

$$1, 2, 3, 3, 4, 6, 10, 11, 17, 18, 22, 24, 29, 31, 38, 45, 48, 49, 50, 52$$



7. Determine the range of the data.
8. Determine the interval width.

9. Create a histogram using the frequency table and interval width in question 8.

10. What was the most common interval of commute times?